# Midterm 2 Review Session

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### Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
```

```
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

# 1 Introduction

This chapter walks through the **most common mistakes** that students made on Midterm 2, which covered survival analysis and Cox proportional hazards models[1]. Each section states a mistake, explains *why* it is wrong, and works through the correct approach.

The exam had two parts:

- **Part 1** (Kaplan-Meier[2] and Nelson-Aalen estimators[3]) used a small data set of survival times: `10, 14, 14+, 15+, 18, 21, 25` months (the `+` marks censored[4] observations).
- **Part 2** (Cox proportional hazards regression) used a model fit to $n = 3{,}142$ men from the Western Collaborative Group Study (WCGS), relating time to incident coronary heart disease (CHD) to baseline cigarette smoking and other covariates.

> **Tip**
>
> The mistakes in this chapter are organized so that the most consequential and most frequent errors come first within each part. If you only have time to review a few things, start at the top of each part.

[1] proportional-hazards-models.qmd#def-ph-model
[2] intro-to-survival-analysis.qmd#def-KM-estimator
[3] intro-to-survival-analysis.qmd#def-na-cuhaz-est
[4] intro-to-survival-analysis.qmd#def-right-censor

# 2 Part 1: Kaplan-Meier and Nelson-Aalen

```
library(survival)
library(dplyr)

surv_data <- tibble(
  time  = c(10, 14, 14, 15, 18, 21, 25),
  death = c(1,  1,  0,  0,  1,  1,  1),
  surv  = Surv(time, death)
)
```

Here is the full table that everyone was trying to reproduce; we will refer back to it throughout this part.

```
KM_est <- survfit(surv ~ 1, type = "kaplan-meier", data = surv_data)

surv_table <-
  KM_est |>
  summary(censored = TRUE, data.frame = TRUE) |>
  as_tibble() |>
  mutate(
    hazard       = n.event / n.risk,
    nonhazard    = 1 - hazard,
    cusum_hazard = cumsum(hazard),
    surv_KM      = cumprod(nonhazard),
    surv_NA      = exp(-cusum_hazard)
  ) |>
  select(time, n.risk, n.event, hazard, surv_KM, cusum_hazard, surv_NA)

surv_table |> pander::pander()
```

Table 1: Kaplan-Meier and Nelson-Aalen calculations for the Part 1 data

| time | n.risk | n.event | hazard | surv_KM | cusum_hazard | surv_NA |
|---|---|---|---|---|---|---|
| 10 | 7 | 1 | 0.1429 | 0.8571 | 0.1429 | 0.8669 |
| 14 | 6 | 1 | 0.1667 | 0.7143 | 0.3095 | 0.7338 |
| 15 | 4 | 0 | 0 | 0.7143 | 0.3095 | 0.7338 |
| 18 | 3 | 1 | 0.3333 | 0.4762 | 0.6429 | 0.5258 |
| 21 | 2 | 1 | 0.5 | 0.2381 | 1.143 | 0.3189 |
| 25 | 1 | 1 | 1 | 0 | 2.143 | 0.1173 |

## 2.1 The risk set shrinks by the number of subjects who *leave*

*Exam reference:* ***Exercise 1.1*** *(6 points) — compute the KM and Nelson-Aalen estimates of* $S(t)$.

> ⚠ Common mistake
>
> Reducing the number at risk[a] by **one** at every row, regardless of how many subjects actually left the study. With this mistake, the number at risk after the two censored[b] subjects (at $t = 14^+$ and $t = 15^+$) was computed as 5 instead of 4.

[a] intro-to-survival-analysis.qmd#def-number-at-risk
[b] intro-to-survival-analysis.qmd#def-right-censor

Solution

*Solution.* The number at risk $n_j$ is the number of subjects still under observation *just before* time $t_j$ — that is, everyone who has neither had the event nor been censored yet.
Track the cohort of 7:

| time | what happens | number at risk *before* this time |
|---|---|---|
| 10 | 1 death | 7 |
| 14 | 1 death **and** 1 censored | 6 |
| 15 | 1 censored | 4 |
| 18 | 1 death | 3 |
| 21 | 1 death | 2 |
| 25 | 1 death | 1 |

At $t = 14$, one subject dies *and* a different subject is censored (the `14+`), so **two** subjects leave the risk set[a]; the count drops from 6 to 4, not from 6 to 5. Every subject who has an event **or** is censored is removed from all later risk sets.

[a] intro-to-survival-analysis.qmd#def-risk-set

## 2.2 Nelson-Aalen estimates the *cumulative hazard* first, then exponentiates

*Exam reference:* ***Exercise 1.1*** *(6 points) — compute the KM and Nelson-Aalen estimates of* $S(t)$.

Common mistake

Two versions of this error were common:

1. Computing a running sum of arbitrary quantities (e.g. summing survival probabilities, or summing $1 - \hat{\lambda}_j$) instead of summing the hazards[a] $\hat{\lambda}_j = d_j/n_j$.
2. Computing the cumulative hazard[b] correctly but then forgetting the final step, $\hat{\mathrm{S}}_{\mathrm{NA}}(t) = \exp\{-\hat{\Lambda}(t)\}$, and reporting the cumulative hazard itself (or the KM product) as the Nelson-Aalen survival estimate.

[a] intro-to-survival-analysis.qmd#def-hazard
[b] intro-to-survival-analysis.qmd#def-cuhaz

Solution

*Solution.* The Nelson-Aalen estimator[a] builds the **cumulative hazard** by adding up the instantaneous hazard contributions at each event time:

$$\hat{\Lambda}(t) = \sum_{j:t_j \le t} \frac{d_j}{n_j}$$

and then converts it to a survival estimate using $\mathrm{S}(t) = \exp\{-\Lambda(t)\}$:

$$\hat{\mathrm{S}}_{\mathrm{NA}}(t) = \exp\{-\hat{\Lambda}(t)\}$$

For example, at $t = 10$:

$$\hat{\Lambda}(10) = \frac{1}{7} = 0.143, \qquad \hat{\mathrm{S}}_{\mathrm{NA}}(10) = \exp\{-0.143\} = 0.867$$

Contrast this with Kaplan-Meier[b], which multiplies conditional survival probabilities[c]:

$$\hat{\mathrm{S}}_{\mathrm{KM}}(t) = \prod_{j:t_j \le t} \left(1 - \frac{d_j}{n_j}\right), \qquad \hat{\mathrm{S}}_{\mathrm{KM}}(10) = 1 - \tfrac{1}{7} = 0.857$$

The two estimators are close but **not** identical, and $\hat{S}_{NA}(t) \geq \hat{S}_{KM}(t)$ always. Reporting the same numbers for both, or reporting $\hat{\Lambda}$ where $\hat{S}_{NA}$ was asked for, loses credit.

[a] intro-to-survival-analysis.qmd#def-na-cuhaz-est
[b] intro-to-survival-analysis.qmd#def-KM-estimator
[c] intro-to-survival-analysis.qmd#def-kappa

## 2.3 Survival estimates are *step functions*: use the most recent event time

*Exam reference:* ***Exercise 1.3*** *(1 point) — the KM and NA survival estimates for $t = 17$ months.*

**Common mistake**

For $\hat{S}(17)$, looking at the row for $t = 18$ (the next event *after* 17), or interpolating between rows. Some answers gave only a verbal description ("it's flat there") without the numeric value.

**Solution**

*Solution.* $\hat{S}(t)$ is a **right-continuous step function**: it only changes at observed event times and stays flat in between. To evaluate it at $t = 17$, find the **most recent event time at or before 17**, which is $t = 14$ (the subsequent $t = 15^+$ is a censoring, not an event, so the curve does not step there). So:

$$\hat{S}_{KM}(17) = \hat{S}_{KM}(14) = 0.714, \qquad \hat{S}_{NA}(17) = \hat{S}_{NA}(14) = 0.734$$

Give the actual numbers, not just "it's flat."

## 2.4 Median survival time is read off the curve, not modeled

*Exam reference:* ***Exercise 1.4*** *(1 point) — the KM and NA estimates of median survival time*[5].

**Common mistake**

Estimating the median by fitting an exponential[a] model ($\hat{\lambda}$ = events/total follow-up, then $\hat{E}[T] = 1/\hat{\lambda}$). That computes an exponential-model **mean**, not the nonparametric **median**, and answers a different question. A second error was reading the median off the wrong curve or the wrong row.

[a] https://morrison-lab.github.io/pds/random-variables.html#def-exponential

**Solution**

*Solution.* The median survival time is the smallest $t$ at which the estimated survival curve drops to (or below) 0.5:

$$\hat{t}_{median} = \min\{t : \hat{S}(t) \leq 0.5\}$$

Reading from Table 1:

- $\hat{S}_{KM}(t)$ first reaches $\leq 0.5$ at $t = 18$ (where it drops to 0.476), so the KM median is **18 months**.
- $\hat{S}_{NA}(t)$ first reaches $\leq 0.5$ at $t = 21$ (where it drops to 0.319), so the NA median is **21 months**.

We can confirm this in R:

[5] intro-to-survival-analysis.qmd#def-median-survival

```r
quantile(KM_est, p = 0.5)$quantile
#> 50
#> 18
```

No distributional assumption (exponential or otherwise) is used: the Kaplan–Meier estimator[a] is a nonparametric read-off of the estimated curve.

[a] intro-to-survival-analysis.qmd#def-KM-estimator

### 2.5 How censored times are used: denominator yes, numerator no

*Exam reference:* ***Exercise 1.5*** *(2 points) — describe how the censored time(s) are utilized in the Kaplan-Meier estimate.*

⚠ Common mistake

Describing only the *symptom* ("the curve doesn't drop at a censoring time") without the *mechanism*, or implying that censored subjects are simply dropped from the analysis entirely.

Solution

*Solution.* A censored subject contributes to the **risk-set**[a] **denominators** ($n_j$) for every event time up to and including their censoring time, because we know they survived event-free until then. They **never** contribute to an event **numerator** ($d_j$), because no event was observed for them.
In this example, subjects #3 and #4 (censored at $14^+$ and $15^+$) are counted in the number at risk at $t = 10$ and $t = 14$, which is why those denominators are 7 and 6. After their censoring times they drop out of all later risk sets. This mechanism is exactly what lets Kaplan-Meier[b] use partial information from censored subjects instead of discarding them.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] intro-to-survival-analysis.qmd#def-KM-estimator

## 3 Part 2: Cox proportional hazards models

For reference, here is the fitted model from the exam. The primary exposure is baseline smoking category (nonsmoker = reference, light smoker $L = 1$, heavy smoker $H = 1$), adjusting for age in decades ($A$), behavior pattern ($P$), overweight ($B$), and high cholesterol ($C$).

Table 2: Cox PH regression for the WCGS CHD data

| Characteristic | $\log(\widehat{HR})$ | $\widehat{HR}$ | 95% CI for HR | $p$ |
|---|---|---|---|---|
| nonsmoker (ref.) | – | – | – | – |
| light smoker ($L$) | 0.36 | 1.43 | (1.05, 1.95) | 0.023 |
| heavy smoker ($H$) | 0.78 | 2.18 | (1.63, 2.91) | $<$ 0.001 |
| age per decade ($A$) | 0.66 | 1.94 | (1.56, 2.40) | $<$ 0.001 |
| behavior type A ($P$) | 0.73 | 2.06 | (1.58, 2.70) | $<$ 0.001 |
| overweight ($B$) | 0.30 | 1.34 | (1.05, 1.72) | 0.019 |
| high cholesterol ($C$) | 0.75 | 2.12 | (1.65, 2.71) | $<$ 0.001 |

The exam also provided the estimated covariance matrix[6] of the coefficient estimates (on the $\log(HR)$

[6] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov-vec-x

scale). The relevant entries are:

$$\mathrm{Var}(\hat{\beta}_L) = 0.024840, \quad \mathrm{Var}(\hat{\beta}_H) = 0.021905, \quad \mathrm{Cov}(\hat{\beta}_L, \hat{\beta}_H) = 0.010504$$

### 3.1 Writing the model: name every assumption and show where it is used

*Exam reference:* ***Exercise 2.1*** *(10 points) — write the mathematical form of the proportional hazards model corresponding to Table 1.*

⚠ Common mistake

Several patterns lost the most points here:

- Treating the question as "interpret the coefficients" and writing only the linear predictor[a], omitting the **likelihood**[b], the **distribution functions**, and the **assumptions**.
- Listing assumption *names* without **showing where they are used**, or writing the math without naming the assumption.
- Adding a spurious **"baseline hazard**[c] **is exponential"** assumption. The Cox model leaves the baseline hazard $\lambda_0(t)$ **unspecified**; assuming a parametric form is wrong.
- Omitting one or more of the standalone distribution functions (survival[d], density[e], hazard[f], log-hazard[g], cumulative hazard[h]).

[a] Intro-to-GLMs.qmd#def-linear-predictor
[b] https://morrison-lab.github.io/sds/intro-MLEs.html#def-lik
[c] proportional-hazards-models.qmd#def-base-hazard
[d] intro-to-survival-analysis.qmd#def-surv-fn
[e] https://morrison-lab.github.io/pds/random-variables.html#def-pdf
[f] intro-to-survival-analysis.qmd#def-hazard
[g] proportional-hazards-models.qmd#def-loghaz
[h] intro-to-survival-analysis.qmd#def-cuhaz

Solution

*Solution.* A full-credit answer connects the likelihood to the linear predictor through a chain of named components and assumptions. Using $\tilde{X} = (L, H, A, P, B, C)$:

**Joint likelihood of the data set:**

$$\mathcal{L} \overset{\text{def}}{=} \mathrm{p}(\tilde{Y} = \tilde{y}, \tilde{D} = \tilde{d} \mid \mathbf{X} = \mathbf{x})$$

**Marginal likelihood contribution of observation $i$:**

$$\mathcal{L}_i \overset{\text{def}}{=} \mathrm{p}(Y_i = y_i, D_i = d_i \mid \tilde{X}_i = \tilde{x}_i)$$

***Independent***[a]***-observations assumption*** (used to factor the likelihood):

$$\mathcal{L} = \prod_{i=1}^{n} \mathcal{L}_i$$

***Non-informative censoring***[b] ***assumption*** (used so each contribution reduces to survival and hazard terms): $T_i \perp\!\!\!\perp C_i \mid \tilde{X}_i$, giving

$$\mathcal{L}_i \propto [\mathrm{f}(y_i \mid \tilde{x}_i)]^{d_i} [\mathrm{S}(y_i \mid \tilde{x}_i)]^{1-d_i} = \mathrm{S}(y_i \mid \tilde{x}_i) \cdot [\lambda(y_i \mid \tilde{x}_i)]^{d_i}$$

**Distribution functions** (define each one):

$$\mathrm{S}(t \mid \tilde{x}) \overset{\text{def}}{=} \mathbb{P}(T > t \mid \tilde{X} = \tilde{x}) = \exp\{-\Lambda(t \mid \tilde{x})\}$$

$$\mathrm{f}(t \mid \tilde{x}) \overset{\text{def}}{=} \lambda(t \mid \tilde{x})\ \mathrm{S}(t \mid \tilde{x})$$

$$\lambda(t \mid \tilde{x}) \overset{\text{def}}{=} \mathrm{p}(T = t \mid T \geq t, \tilde{X} = \tilde{x}) = \frac{\mathrm{f}(t \mid \tilde{x})}{\mathrm{S}(t \mid \tilde{x})}$$

$$\Lambda(t \mid \tilde{x}) \overset{\text{def}}{=} \int_0^t \lambda(u \mid \tilde{x})\, du = -\log \mathrm{S}(t \mid \tilde{x})$$

$$\eta(t \mid \tilde{x}) \overset{\text{def}}{=} \log \lambda(t \mid \tilde{x})$$

***Proportional-hazards*[c] *assumption*** (used to split the hazard into a baseline that depends only on time and a factor that depends only on covariates):

$$\lambda(t \mid \tilde{x}) = \lambda_0(t) \cdot \theta(\tilde{x})$$

where $\lambda_0(t)$ is the **unspecified** baseline hazard.
***Logarithmic-link*[d] *assumption*** (used to make the covariate factor a function of a linear predictor):

$$\eta(t \mid \tilde{x}) = \eta_0(t) + \Delta\eta(\tilde{x}), \qquad \theta(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$$

***Linear functional-form assumption*** (used to write the covariate term as a linear combination[e]):

$$\Delta\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} = \beta_L\, l + \beta_H\, h + \beta_A\, a + \beta_P\, p + \beta_B\, b + \beta_C\, c$$

Notice that the baseline hazard $\lambda_0(t)$ is carried along symbolically the whole time — we never assume a shape for it.

[a] https://morrison-lab.github.io/pds/independence.html#def-indpt
[b] intro-to-survival-analysis.qmd#def-noninformative-censoring
[c] proportional-hazards-models.qmd#def-pha
[d] Intro-to-GLMs.qmd#sec-why-link
[e] https://morrison-lab.github.io/mds/linear-algebra-vectors.html#def-linear-combination

## 3.2 Interpreting a hazard ratio: magnitude, reference, adjustment, *and* significance

*Exam reference:* ***Exercise 2.2*** *(4 points) — summarize how baseline smoking category is associated with hazard of incident CHD.*

⚠ Common mistake

- Reporting the direction of the effect but omitting the **magnitude** ("X% higher/lower hazard") or the **reference group**.
- Forgetting to say the estimate is **adjusted for / holding constant** the other covariates.
- Omitting **statistical significance**[a] (whether the 95% CI[b] excludes 1, or $p < 0.05$).
- Calling the hazard ratio[c] a **risk** or an **odds ratio**[d]. It is a ratio of *hazards*, not of risks or odds[e].

[a] https://morrison-lab.github.io/sds/inference.html#def-statistically-significant
[b] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[c] intro-to-survival-analysis.qmd#def-hazard-ratio
[d] binary-outcome-associations.qmd#def-OR
[e] binary-outcome-associations.qmd#def-odds

Solution

*Solution.* A complete interpretation has four parts: magnitude, reference group[a], adjustment, and significance. For the smoking effect in Table 2:

> Adjusting for age, behavior pattern, overweight, and cholesterol, **light smokers** had an estimated hazard of incident CHD about **43% higher** than **nonsmokers** ($\widehat{HR} = 1.43$); this hazard ratio is statistically significant at the 0.05 level, since the 95% CI $(1.05, 1.95)$ excludes 1 ($p = 0.023$).
> **Heavy smokers** had an estimated hazard about **118% higher** (roughly double; $\widehat{HR} = 2.18$) than nonsmokers, also adjusting for the other covariates; this hazard

ratio is highly significant ($p < 0.001$, CI $(1.63, 2.91)$).
Note "43% higher" comes from $1.43 - 1 = 0.43$, and "118% higher" from $2.18 - 1 = 1.18$. Interpret each smoking level **separately** against the reference, rather than lumping them together.

[a] https://morrison-lab.github.io/sds/data.html#def-reference-level

### 3.3 A hazard ratio for a non-unit change: exponentiate

*Exam reference:* ***Exercise 2.3*** *(2 points) — the hazard ratio associated with a 7.5-year increase in age.*

⚠ Common mistake

For the hazard ratio[a] associated with a **7.5-year** increase in age (with the table reporting the HR per **decade**), the errors were:

- Multiplying: $1.94 \times 0.75 = 1.46$. **Wrong** — the HR for a multi-unit change is *not* linear in the HR.
- Arithmetic slips that produced answers like 2.08.

[a] intro-to-survival-analysis.qmd#def-hazard-ratio

Solution

*Solution.* On the **log** scale, the effect is linear: a change of $c$ units multiplies the log-hazard[a] by $c \cdot \beta$. So the hazard ratio for a $c$-unit change is

$$HR(c) = \exp\{c\,\beta\} = (\exp\{\beta\})^c = HR^c$$

A 7.5-year increase is $c = 0.75$ decades, and the per-decade $\widehat{HR} = 1.94$ (i.e. $\hat{\beta}_A = \log 1.94 = 0.66$):

$$\widehat{HR}(0.75) = 1.94^{0.75} = \exp\{0.75 \times 0.66\} = \exp\{0.495\} = 1.64$$

So a 7.5-year-older man has about **64% higher** estimated hazard of CHD, all else equal. The operation is **exponentiation** ($HR^{0.75}$), not multiplication.

[a] proportional-hazards-models.qmd#def-loghaz

### 3.4 Comparing two coefficients: variance of a *difference*

*Exam reference:* ***Exercise 2.4*** *(2 points) — test whether the hazard of incident CHD differs between heavy and light smokers; compute the z-statistic*[7] *and two-sided p-value*[8]*.*

⚠ Common mistake

This mistake was the **single most consequential Part 2 error**. To test $H_0 : \beta_H = \beta_L$, the standard error[a] of $\hat{\beta}_H - \hat{\beta}_L$ was computed incorrectly in several ways:

- Using a **single covariance**[b] **entry** $\text{Cov}(\hat{\beta}_L, \hat{\beta}_H)$ as the variance[c].
- **Adding** 2 Cov instead of **subtracting** it.
- Plugging in the **hazard ratios** (2.18, 1.43) instead of the **coefficients** (0.78, 0.36) in the numerator.
- Guessing a value for $z$ because the variance-of-a-difference formula[d] was not on the formula sheet.

[a] https://morrison-lab.github.io/sds/estimation.html#def-SE

[7] https://morrison-lab.github.io/sds/inference.html#def-test-statistic
[8] https://morrison-lab.github.io/sds/inference.html#def-p-value

[b] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov
[c] https://morrison-lab.github.io/pds/variance-covariance.html#def-variance
[d] https://morrison-lab.github.io/pds/variance-covariance.html#thm-var-lincom

Solution

*Solution.* Work on the $\log(HR)$ (coefficient) scale, where the estimates are approximately normal[a]. The point estimate of the difference is

$$\widehat{\Delta} = \hat{\beta}_H - \hat{\beta}_L = 0.78 - 0.36 = 0.42$$

The variance of a **difference** of two estimates uses **all three** relevant entries of the covariance matrix[b]:

$$\mathrm{Var}(\hat{\beta}_H - \hat{\beta}_L) = \mathrm{Var}(\hat{\beta}_H) + \mathrm{Var}(\hat{\beta}_L) - 2\ \mathrm{Cov}(\hat{\beta}_H, \hat{\beta}_L)$$

The $-2$ Cov term is essential — its sign is **minus** for a difference. Plugging in:

```
var_H <- 0.021905
var_L <- 0.024840
cov_HL <- 0.010504

diff <- 0.78 - 0.36
var_diff <- var_H + var_L - 2 * cov_HL
se_diff <- sqrt(var_diff)

z <- diff / se_diff
p_value <- 2 * pnorm(-abs(z))

c(diff = diff, se = se_diff, z = z, p_value = p_value) |> round(4)
#>    diff      se       z p_value
#>  0.4200  0.1604  2.6180  0.0088
```

So $z = 0.42/0.160 = 2.62$, two-sided $p = 0.0088$. We **reject** $H_0$: the hazard of CHD differs significantly[c] between heavy and light smokers, holding the other covariates constant.

**Note**

Compare with the **naive** standard error that ignores the covariance, $\sqrt{\mathrm{Var}(\hat{\beta}_H) + \mathrm{Var}(\hat{\beta}_L)} = 0.216$. Because $\hat{\beta}_H$ and $\hat{\beta}_L$ are *positively* correlated[d], subtracting 2 Cov makes the correct SE (0.160) **smaller**, which sharpens the test. Using a single covariance cell as the variance gave a wildly wrong SE and an absurd $z$.

[a] https://morrison-lab.github.io/pds/random-variables.html#def-normal
[b] https://morrison-lab.github.io/pds/variance-covariance.html#def-cov-vec-x
[c] https://morrison-lab.github.io/sds/inference.html#def-statistically-significant
[d] https://morrison-lab.github.io/pds/variance-covariance.html#def-correlation

## 3.5 A confidence interval for a hazard ratio: build it on the log scale, then exponentiate

*Exam reference:* ***Exercise 2.5*** *(2 points) — a 95% confidence interval*[9] *for the hazard ratio*[10] *comparing heavy to light smokers.*

[9] https://morrison-lab.github.io/sds/inference.html#def-confidence-interval
[10] intro-to-survival-analysis.qmd#def-hazard-ratio

⚠ Common mistake

- Forming the interval directly on the HR scale, $\widehat{HR} \pm 1.96 \cdot \text{SE}$ (e.g. $1.52 \pm 0.31$), instead of on the log scale.
- Building the log-scale interval correctly but **never exponentiating**, then reporting a log-scale interval as if it were the HR (sometimes even declaring "significant" while the reported interval contained 1).
- Reusing the wrong SE from the Wald test[a] in Section 3.4.

[a] https://morrison-lab.github.io/sds/intro-MLEs.html#def-wald-test

Solution

*Solution.* A hazard ratio is positive and its sampling distribution is skewed, so the symmetric normal approximation is applied to $\log(HR)$, and the endpoints are **then** exponentiated. For the heavy-vs-light comparison ($\widehat{\Delta} = 0.42$, SE $= 0.160$):

$$\text{log-scale CI:} \quad 0.42 \pm 1.96 \times 0.160 = (0.106,\ 0.734)$$

$$\text{HR CI:} \quad (e^{0.106},\ e^{0.734}) = (1.11,\ 2.08)$$

```
est <- 0.42
se  <- sqrt(0.021905 + 0.024840 - 2 * 0.010504)
exp(est + c(-1, 1) * 1.96 * se) |> round(2)
#> [1] 1.11 2.08
```

The point estimate is $\widehat{HR} = e^{0.42} = 1.52$, and the 95% CI $(1.11, 2.08)$ **excludes 1**, consistent with the significant Wald test in Section 3.4. Always exponentiate the endpoints, and check that your CI and your hypothesis test agree.

# 4 Summary of the most common errors

| # | Exam Q | Topic | The fix |
|---|---|---|---|
| 1 | 1.1 | Risk set after censoring (Section 2.1) | Remove **everyone** who has an event *or* is censored from later risk sets. |
| 2 | 1.1 | Nelson-Aalen (Section 2.2) | Sum hazards $d_j/n_j$, then $\hat{\text{S}}_{\text{NA}} = \exp\{-\hat{\Lambda}\}$. |
| 3 | 1.3 | Reading $\hat{S}(t)$ (Section 2.3) | Step function: use the most recent event time $\le t$; give the number. |
| 4 | 1.4 | Median survival (Section 2.4) | First $t$ with $\hat{S}(t) \le 0.5$ — not a mean, not an exponential fit. |
| 5 | 1.5 | Role of censoring (Section 2.5) | Denominator until censoring; never the event numerator. |
| 6 | 2.1 | Writing the model (Section 3.1) | Name **and** apply every assumption; baseline hazard stays unspecified. |
| 7 | 2.2 | Interpreting an HR (Section 3.2) | Magnitude + reference + adjustment + significance; HR $\neq$ risk/odds. |
| 8 | 2.3 | HR for a non-unit change (Section 3.3) | $HR^c = \exp\{c\beta\}$ — exponentiate, don't multiply. |

| # | Exam Q | Topic | The fix |
|---|---|---|---|
| 9 | 2.4 | Comparing two coefficients (Section 3.4) | $\text{Var}(\hat{\beta}_H - \hat{\beta}_L) = \text{Var}_H + \text{Var}_L - 2\ \text{Cov}.$ |
| 10 | 2.5 | CI for an HR (Section 3.5) | Build on the log scale, then exponentiate the endpoints. |